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VECTOR SPACE PROBLEM SHEET

This paper asks you to prove some well known results. Although the algebra is easy the proofs are not entirely straightforward. There are marks assigned to the readability of the solution and also how well laid out and explained the steps you make are. (A good proof needs to be easy to follow: you need not comment on trivial algebra, but there should not be steps that are difficult to follow).

This looks very mathematical, but it helps to develop the tools and language that is used to describe machine learning.

1 An inner product $\langle x, y \rangle$ between vectors in a vector space $\mathcal{V}$ satisfies the following properties

- (a) $\langle x, x \rangle \geq 0$ for all $x \in \mathcal{V}$
- (b) $\langle x, x \rangle = 0$ if and only if $x = 0$
- (c) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$
- (d) $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$
- (e) $\langle x, y \rangle = \langle y, x \rangle$

The question explores properties of inner products.

(a) Consider the quadratic

$$q(t) = \langle \mathbf{x} + t\mathbf{y}, \mathbf{x} + t\mathbf{y} \rangle$$

By definition 1 of an inner product $q(t)$ must be non-negative and will only be 0 when $\mathbf{x} + t\mathbf{y} = \mathbf{0}$.

Expand $q(t)$ in the form $q(t) = At^2 + 2Bt + C$ to find A , B and C . For $q(t)$ to not change sign its roots (values of t such that $q(t) = 0$) must be complex (i.e. have an imaginary part), or possible have a double root if there is a value of t such that $q(t) = 0$. Use the standard solutions to a quadratic of the form $q(t) = At^2 + 2Bt + C$ to show that for our $q(t)$ to never become negative implies

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

This is the famous Cauchy-Schwarz inequality written in a very general form. [10 marks]

[illegible]

- (b) From an inner product we can define a norm $\|x\| = \sqrt{\langle x, x \rangle}$. This clearly satisfies non-negativity and linearity. The only non-trivial property to show is that this norm satisfies the triangular inequality. Expand out $\|x + y\|$ and hence show that

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\langle x, y \rangle$$

Then use the Cauchy-Schwarz inequality to prove the triangular inequality.

[10 marks]

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10

End of question 1

(a) $\frac{\quad}{10}$	(b) $\frac{\quad}{10}$	Total	$\frac{\quad}{20}$

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2 Random variables, X , Y , etc. form a vector space (i.e. they satisfy properties such as closure under addition and scalar multiplication). Furthermore we can define an inner product between random variables as

$$\langle X, Y \rangle = \mathbb{E}[XY]$$

(i.e. the expectation of the random variable $Z = XY$)

(a) Use the Cauchy-Schwarz inequality to show that

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X)\text{Var}(Y)$$

hence show that the Pearson correlation is between -1 and 1. [10 marks]

10

(b) Show that for vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ that the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i$$

satisfies the condition of an inner product provided $w_i > 0$ for all i . Write down the norm and distance induced by this inner product and provide an interpretation of what this distance means.

[10 marks]

10

End of question 2

(a) $\frac{\quad}{10}$	(b) $\frac{\quad}{10}$	Total $\frac{\quad}{20}$
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3 In the previous question you showed that $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i$ is an inner product provided every weight w_i is positive. Functions form a vector space in just the same way (we can add them and multiply them by scalars), and if we replace the sum over components by an integral over x we get

$$\langle f, g \rangle = \int_{-\infty}^{\infty} w(x) f(x) g(x) dx$$

which, by the same argument, is an inner product provided the *weight function* $w(x)$ is positive.

In this question we choose $w(x)$ to be the density of a standard normal deviate

$$w(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

so that $\langle f, g \rangle = \mathbb{E}[f(X)g(X)]$ where $X \sim \mathcal{N}(0,1)$. That is, this is exactly the inner product of the previous question, applied to functions of a random variable.

The monomials $f_i(x) = x^i$ span the polynomials, but they are not orthogonal. We can orthogonalise them with the Gram–Schmidt procedure: having found $p_0, p_1, \dots, p_{n-1}$ we subtract from f_n its component along each of them,

$$p_n(x) = f_n(x) - \sum_{i=0}^{n-1} \frac{\langle f_n, p_i \rangle}{\langle p_i, p_i \rangle} p_i(x), \quad p_0(x) = 1.$$

Orthogonal polynomials are useful because they make approximating a function easy. If we want the best polynomial approximation to some function f we can compute each coefficient separately with a single inner product, and if we later decide to add a higher order term none of the coefficients we have already found change. Using the monomials instead we would have to solve a set of simultaneous equations, and start again from scratch each time we added a term.

You will need the integrals below, which you are not expected to compute. (They are the moments of a standard normal deviate. All the odd moments vanish because $w(x)x^n$ is an odd function.)

$\mathbb{E}[X^n] = \int_{-\infty}^{\infty} w(x)x^n dx$	n	0	1	2	3	4	5	6
	1	0	1	0	3	0	15	

(c) Find $p_3(x)$.

[5 marks]

5

